

Assignment 5

Laws of Expectation, Properties of Variance, MGF, Moment of Higher Order

1. If a random variable X has the following probability distribution, find $E(3X^2-2X+5)$ (Ans:10.75)

X	-2	-1	0	1	2
$P(X=x)$	0.20	0.30	0.15	0.25	0.10

2. If a random variable X has the probability distribution given above, find the probability distribution of $Y=X^2$ and find $E(Y)$. (Ans: 1.75)
3. A random variable X has the probability distribution $P(X=x) = 1/2^x$, $x=1,2,3,\dots$. A new random variable Y is defined by $Y=1$ if x is divisible by 2 and $Y=0$ otherwise. Find the probability distribution of Y and $E(Y)$
[Ans: $Y: 0 \quad 1$] $[E(Y)=1/3]$
 $P(Y=y): 2/3 \quad 1/3$
4. A random variable X has the following pmf
 $P(X=x) = x/15$, $x = 1,2,3,4,5$.
Find $E(X)$, $E(2X^2+1)$, $E(3X+8)$ (Ans: 11/3,31,19)
5. Suppose that X is R.V with $E(X) = 10$ and $V(X) = 25$. Find the positive values of a and b such that $Y=aX-b$ has expectation 0 and variance 1. (Ans: 1/5,2)
6. If $E(X)=2$ and $V(X) = 5$, find $E(3X+2)$ and $V(3X+2)$. (Ans: 8,45)
7. Find $V(X+2)$ and $V(3X+2)$. (Ans: $V(X)$, $9V(X)$)
8. If X_1 has mean 5 and variance 5, X_2 has mean -2 and variance 3 and if X_1 and X_2 are independent, find (i) $E(X_1+X_2)$, $V(X_1+X_2)$ (ii) $E(X_1-X_2)$, $V(X_1-X_2)$ (iii) $E(2X_1+3X_2-5)$, $V(2X_1+3X_2-5)$. Ans: (i) 3,8 (ii) 7,8 (iii) -1,47
9. Find the expectation of (i) the sum, (ii) the product of the number of points on the throw of n dice. [Ans: $7n/2$, $(7/2)^n$]
10. The joint probability density function of (X,Y) is given by

$$f(x,y) = \begin{cases} \frac{xy}{8}, & 0 < x < 2, 1 < y < 3 \\ 0, & \text{otherwise} \end{cases}$$

Find (i) $E(X)$, (ii) $E(Y)$, (iii) $E(2X+3Y)$. (Ans: 4/3,13/6,55/6)

11. Let X,Y be the numbers obtained on two dice when thrown. Find the expectations of (i) X (ii) $X-Y$ (iii) $2X-3Y$ [Ans: 7/2,0,-7/2]
12. Find the mgf of a random variable X if the r th moment about the origin is given by $\mu_r' = r!$
13. A random variable X has probability density function

$$f(x) = \begin{cases} ke^{-kx}, & x > 0, k > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find mgf and hence the mean and variance (Ans: $1/k, 1/k^2$)

14. The probability density function of a random variable is given by

$$f(x) = \begin{cases} 1/k, & 0 < x < k \\ 0, & \text{elsewhere} \end{cases}$$

Find mgf, r^{th} moment and hence the mean and variance (Ans: $k/2, k^2/12$)

15. A random variable X has probability density function

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find mgf and μ_r' , mean and variance (Ans: $1/3, 1/12$)